



A Plagiarism Detector in Learning Management System for UTHM

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CHAPTER THREE

METHODOLOGY

This chapter explain the use of prototype model in this project and the activities that had been carried out in each phase. Section 3.1 shows that this project will use an Agile Model Methodology for System Development. The Agile Model Methodology creates the entire system with multiple cycles. The Agile Model Methodology allows each model to be enhanced through constant feedback from lecturers and students. Section 3.2 showed a highly structured development workflow is used, with 5 distinct phases, Planning, Analysis, Design, Implementations, Testing, and specific activities and deliverables to create a user-friendly solution.

3.1 Agile Model

The Agile Model is an iterative development process that creates prototype system versions that can be further enhanced through multiple iterations[1] Continuous delivery of user feedback helps to implement ongoing improvements to each new version of the prototype. The Agile approach emphasizes user satisfaction, collaborative processes, and flexibility in producing a final product that continues to improve over time by using the multiple refinements from iterations throughout the entire life cycle of the product. The Agile Model contains the following phases, Planning, Analysis, Design, Implementation, Testing, Maintenance. Each phase is developed from the last, and iterations are utilized between phases to allow for continuous sharing between phases until the prototypes develop into the final system.

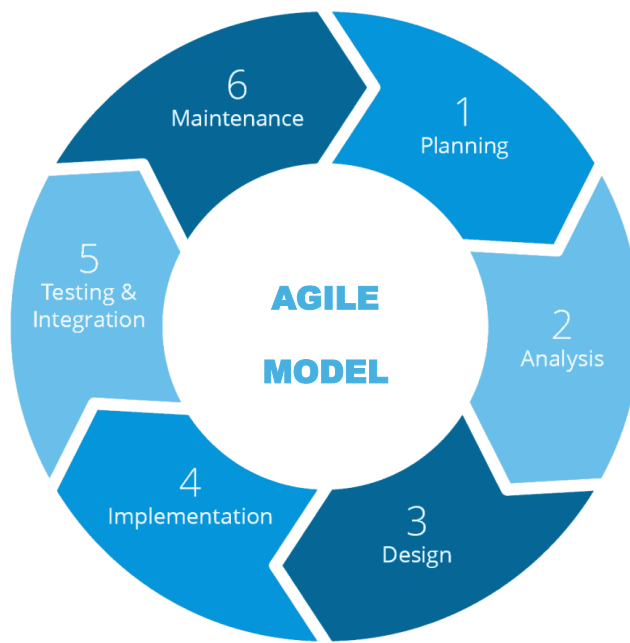


Diagram 3.1: Agile Model Phases

3.1.1 Planning Phase

During the planning phase of the project, the information was gathered on the project goals, objectives, and requirements through discussions with stakeholders including, lecturers, students, and UTHM IT portal team (Pusat Teknologi Maklumat UTHM). The planning phase defines the requirements for the plagiarism detection module. For example, the ability to upload documents, check for similarities, manage users, and create reports. The project scope, project constraints, timeline and technologies required for the project are also established. The planning phase creates user stories or feature lists to represent what the system will be like from a user's perspective. User stories and lists will form the basis of future development cycles. The planning phase will help ensure that the development process is understandable on what will need to be developed prior to moving on.

3.1.2 Analysis Phase

To successfully design and develop a plagiarism detection in a Learning Management System, it is important to first conduct an analysis phase of each functional and technical requirement. The Analysis phase is necessary to determine the functional and technical requirements for a plagiarism detection module, and to prepare the system to

meet the specific needs of its users, including students and lecturers. The analysis phase identifies student needs, including the ability for students to upload assignments with ease and receive notifications when their assignment has been submitted. Lecturer needs include automated similarity verification, detailed similarity reports, and ease of access to student documents. The analysis phase also determines the types of algorithms that will be used to process, compare, and validate submitted documents and defines how submitted documents will be processed, stored, and compared.

The analysis of the current challenges faced by the AUTHOR as well as the lack of an integrated plagiarism detection system will provide the necessary data to design a new system to meet these needs and specifications. The Analysis phase will also include all technical limitations related to the implementation of a plagiarism detection module, such as the accuracy of algorithms, server loads and response times, in order to develop a complete system that meets all technical requirements. In addition, at the end of the analysis phase a detailed list of system requirements and workflow behaviour for the proposed plagiarism detection module will be available, which will assist in the design and development of the plagiarism detection module.

3.1.3 Design Phase

All components of the system will be organised as part of creating a system design that ensures proper functionality, and ease of use for both lecturers and students. In order to achieve this goal, it was essential to define key elements of the system, define how the plagiarism detection module interacts with the submission process, and provide the user interface layout so that the user interface feels easy to use. Additionally, it was necessary to determine how the system will perform other essential functions, such as file upload capabilities, generating and providing similarity reports, and sending alerts/notifications to users. The database structure will also be developed over time using Entity-Relationship Diagrams (ERD) and Data Flow Diagram (DFD). The development of user interface layouts using Figma will provide clear visualisation and interaction opportunities prior to the beginning of development, as well as allow for ease of collaboration and adaptability based on feedback from both lecturers and students[2] .

3.1.4 Implementation Phase

The Agile development of the system is in phases with an iterative approach following a sprint model to break down each of the functions into smaller segments. The four major incremental system functions are user login, document upload, similarity comparison, and report creation. Each segment will be constructed, tested, and enhanced during its respective sprint until it has reached its final state for production use. Plagiarism detection algorithms will also undergo the same process and be integrated into the workflow for automatic evaluation of all documents uploaded.

Backend functionality will be programmed in PHP and Python, and the underlying data will be stored in a MySQL database. Frontend functionality will be developed in HTML, CSS, and JavaScript. At the end of each module, stakeholders will be able to see how the module operates and provide feedback. Recommended changes will be made before progress continues on the next sprint.

3.1.5 Testing Phase

The testing process is designed to verify that the system performs in accordance with the requirements specified and that it works correctly and efficiently. The first stage of testing is Unit Testing. During this stage each Module (e.g., File Uploading, Similarity Check Calculation, Report Generation) is tested separately to verify it is functioning as expected. The next stage of testing is System Integration Testing. During System Integration Testing, modules are tested together as a complete system and the flow of data between Modules is verified as being error-free. The final stage of testing is User Acceptance Testing (UAT). During UAT, Lecturers and Students test the system to assure themselves that it is easy to use, performs as intended, and satisfies the needs of its users. Each of the above tests contributes to creating a reliable, usable, and deployable system [3].

3.1.6 Maintenance Phase

During the Maintenance Phase, the plagiarism detection tool is also supported via the continuation of development. This includes fixing any bugs that have been reported by users as well as enhancing detection of plagiarism through updates to the existing algorithms and enhancing overall performance. New functionalities, interface improvements, or security enhancements based on feedback from lecturers and

students to the Plagiarism Detection Tool will be implemented during the maintenance phase. Improvements to maintain an acceptable level of compliance to academic standards will also be included.

3.2 System Development Workflow

There are a total of six phases from the prototype model. As shown in Table 3.1, each phase has its assignment and output that needs to be produced during the entire project development.

Table 3.1: Software development activities and their task

Phase	Tasks	Outputs
Planning	• Define project objectives and scope • Identify users (lecturer & student) • Determine features and constraints • Develop project timeline and resource plan	• Project scope definition • Preliminary timeline
Analysis	• Gather system requirements from lecturer and students • Identify functional and non-functional requirements • Prepare use-case descriptions and workflow scenarios	• Software Requirements Specification (SRS) • List of system requirements • Develop Gantt chart
Design	• Develop system architecture and workflow • Design user interface layout and mockups	• UI mockups/layouts • System architecture design • Database schema and DFD

	• Plan database schema and data relationships	• Entity Relationship Diagram
Implementation	• Develop UI using chosen technology • Code system modules (submission, evaluation, feedback, etc.) • Integrate front-end and database • Conduct internal debugging	• Working prototype • Functional module integration • Source code files
Testing	• Conduct functional testing of each feature • Perform user acceptance testing with lecturer and students • Identify and log bugs/issues	• Test case results • UAT feedback forms • List of bugs and improvement items
Maintenance	• Collect detailed feedback from lecturers and students • Review issues, usability concerns, missing requirements • Refine and update prototype based on findings • Retest after modifications	• Improvement summary • Updated prototype version • Validation report after iteration

3.3 Chapter Summary

The methodology used to design the prototype plagiarism detection system for UTHM is detailed in this chapter. This chapter introduces the iterative approach selected because it enables ongoing improvements based on input from both lecturers and students. Each phase of development is outlined, beginning with the planning phase where objectives and scope are established, then moving on to analysing the functional requirements and user requirements. The design phase provides an overview of the system architecture, interface design, and database schema that were all converted into a live prototype during implementation. Once the prototype was developed, testing was performed to ensure functionality and usability of the system and any problems found were addressed through the structured test and review cycle outlined in the methodology. This methodology allowed for the creation of a practical and user-friendly system that effectively meets the goals of Academic Integrity.

References

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- [3] S. Kulkarni, "The Necessity of System Integration Testing (SIT) and User Acceptance Testing (UAT) in Project Lifecycle," *Interantional Journal of Scientific Research in Engineering and Management*, vol. 08, no. 11, pp. 1–7, 2024, doi: 10.55041/ijrem7673.